

Modeling Report

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With heightened interest from the group in seeing what demographics are more susceptible to telemarketing from banks, the main question for this project arose: what are the impacts of characteristics of an individual, such as their age, marital status, education level, etc., that influence their likelihood of committing to a telemarketing call? More critically, the group hopes to find a few dominant characteristics that may hopefully explain most of the variance from our dependent variable through methods of linear modeling. One real-world insight that can potentially be learned is how to identify individuals who are likely to commit to fixed-income products out of a pool of potential candidates. And more importantly, this could give insight into the fiscal tendencies of various demographics through our modeling.

The primary focus of the model should be prediction focused, as the identification of individuals that are susceptible to bank telemarketing is not only useful for bankers but also for individuals to protect their own privacy. However, modeling this banking data could also be helpful in dimensionality reduction. There are many factors that an individual may have that is useful in financial decision-making, but if the model indicates that only a few of the features are useful, then perhaps it could increase the interpretability.

In this modeling section, we want to answer the following question: which feature should we include in our GLM model in order to minimize the mean square error of our prediction? We considered LASSO/Ridge Regression to apply penalties to models with redundant features (shrinkage methods), Forward/Backward selection to select a pseudo-optimal model that would hopefully minimize the mean square error without going through the MSE of every single possible model, and ANOVA to determine which independent variable has the greatest impact on the binary response variable y .

Method 1: Model Selection

In this section, we will randomly split the data into a training set and a testing set. Firstly, we will use the training data to find the pseudo-optimal model using forward selection with AIC, backward selection with AIC, forward selection with BIC, and backward selection with BIC. Then, we will compare the cross-validation mean square error of the four models' predictions. The model with the lowest MSE should be the best model out of the four. And finally, we will make sure that the model is the best by giving four different sets of predictions on the testing data using the four different models, and comparing their the MSE of the four sets of predictions. The four different models are shown in the tables below.

```

##(r)
summary(forwardAzc_model)

call:
glm(formula = y_boolean ~ duration + nr.employed + month + poutcome +
  emp.var.rate + cons.price.idx + contact + cons.conf.idx +
  default + pdays + day_of_week + campaign + job + surtbor3m +
  previous, family = "binomial", data = train)

deviance residuals:
      min       1q   median       3q      max
-5.9207 -0.3018 -0.1864 -0.1359  3.3312

coefficients:
              estimate std. error z value Pr(>|z|)
(Intercept) -2.575e+02  4.543e+01 -5.667 1.45e-08 ***
duration     4.648e-03  8.839e-05 52.589 < 2e-16 ***
nr.employed  7.180e-03  3.695e-05 1.943 0.052021 .
monthaug     7.737e-01  1.432e-01  5.401 6.62e-08 ***
monthdec     4.532e-01  2.425e-01  1.869 0.061646 .
monthjul     5.634e-02  1.145e-01  0.492 0.622771 .
monthjun     -5.723e-01  1.496e-01 -3.826 0.000130 ***
monthmar     1.998e+00  1.728e-01 11.563 < 2e-16 ***
monthmay     -4.840e-01  9.832e-02 -4.923 8.52e-07 ***
monthnov     -5.165e-01  1.450e-01 -3.562 0.000367 ***
monthoct     1.515e-01  1.826e-01  0.830 0.406558 .
monthsep     4.089e-01  2.128e-01  1.922 0.054649 .
poutcome nonexistent 3.582e-01  1.132e-01  3.165 0.001551 **
poutcome success    6.942e-01  2.486e-01  2.792 0.005235 **
emp.var.rate -1.774e+00  1.686e-01 -10.522 < 2e-16 ***
cons.price.idx 2.330e+00  2.993e-01  7.785 6.96e-15 ***
contacttelephone -7.347e-01  9.322e-02 -7.882 3.23e-15 ***
cons.conf.idx  3.032e-02  9.258e-03  3.275 0.001057 **
defaultunknown -3.360e-01  7.910e-02 -4.247 2.16e-05 ***
defaultyes     -7.302e+00  1.385e-02 -0.053 0.957950 .
pdays        -1.257e-03  2.552e-04 -4.925 8.44e-07 ***
day_of_weekmon -1.397e-01  7.871e-02 -1.774 0.075984 .
day_of_weekthu  2.342e-02  7.650e-02  0.306 0.759517 .
day_of_weektue  8.014e-02  7.812e-02  1.026 0.304912 .
day_of_weekwed  1.994e-01  7.795e-02  2.558 0.010533 *
campaign       -3.813e-02  1.366e-02 -2.792 0.005243 **
jobblue-collar -2.921e-01  7.839e-02 -3.727 0.000194 ***
jobentrepreneur -1.610e-01  1.469e-01 -1.096 0.273045 .
jobhousemaid   -1.919e-01  1.738e-01 -1.105 0.269374 .
jobmanagement -2.767e-02  9.975e-02 -0.277 0.781465 .
jobretired     1.211e-01  1.002e-01  1.208 0.226919 .
jobself-employed -1.216e-01  1.396e-01 -0.871 0.383641 .
jobservices    -1.558e-01  9.571e-02 -1.628 0.103513 .
jobstudent     1.394e-01  1.218e-01  1.144 0.252649 .
jobtechnician -2.158e-02  7.579e-02 -0.285 0.775834 .
jobunemployed  -1.636e-01  1.533e-01 -1.054 0.291888 .
jobunknown     -2.268e-01  3.105e-01 -0.730 0.465243 .
surtbor3m      2.582e-01  1.543e-01  1.673 0.094287 .
previous       -1.136e-01  7.184e-02 -1.582 0.113703 .
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 20357 on 28831 degrees of freedom
Residual deviance: 12016 on 28793 degrees of freedom
AIC: 12094

Number of Fisher scoring iterations: 10

```

```

##(r)
summary(backwardAzc_model)

call:
glm(formula = y_boolean ~ job + default + contact + month + day_of_week +
  duration + campaign + pdays + previous + poutcome + emp.var.rate +
  cons.price.idx + cons.conf.idx + surtbor3m + nr.employed,
  family = "binomial", data = train)

deviance residuals:
      min       1q   median       3q      max
-5.9207 -0.3018 -0.1864 -0.1359  3.3312

coefficients:
              estimate std. error z value Pr(>|z|)
(Intercept) -2.575e+02  4.543e+01 -5.667 1.45e-08 ***
jobblue-collar -2.921e-01  7.839e-02 -3.727 0.000194 ***
jobentrepreneur -1.610e-01  1.469e-01 -1.096 0.273045 .
jobhousemaid   -1.919e-01  1.738e-01 -1.105 0.269374 .
jobmanagement -2.767e-02  9.975e-02 -0.277 0.781465 .
jobretired     1.211e-01  1.002e-01  1.208 0.226919 .
jobself-employed -1.216e-01  1.396e-01 -0.871 0.383641 .
jobservices    -1.558e-01  9.571e-02 -1.628 0.103513 .
jobstudent     1.394e-01  1.218e-01  1.144 0.252649 .
jobtechnician -2.158e-02  7.579e-02 -0.285 0.775834 .
jobunemployed  -1.636e-01  1.533e-01 -1.054 0.291888 .
jobunknown     -2.268e-01  3.105e-01 -0.730 0.465243 .
surtbor3m      2.582e-01  1.543e-01  1.673 0.094287 .
previous       -1.136e-01  7.184e-02 -1.582 0.113703 .
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 20357 on 28831 degrees of freedom
Residual deviance: 12016 on 28793 degrees of freedom
AIC: 12094

Number of Fisher scoring iterations: 10

```

```

####r)
summary(forwardaicc_model)

call:
glm(formula = y_boolean ~ duration + nr.employed + month + poutcome +
  emp.var.rate + cons.price.idx + contact + cons.conf.idx +
  pdays + default, family = "binomial", data = train)

Deviance Residuals:
    Min       1q   Median       3q      Max
-5.9154  -0.3045  -0.1869  -0.1380   3.2247

Coefficients:
              estimate std. error z value Pr(>|z|)
(Intercept)   -3.011e-02  3.735e-01  -8.061 7.56e-16 ***
duration       4.639e-03  8.798e-05  52.731 < 2e-16 ***
nr.employed    1.181e-02  2.450e-03  4.820 1.44e-06 ***
monthaug       8.602e-01  1.398e-01  6.152 7.66e-10 ***
monthdec       5.962e-01  2.299e-01  2.593 0.009304 **
monthjul       9.696e-02  1.121e-01  0.865 0.387031
monthjun       -5.933e-01  1.480e-01  -4.008 6.13e-05 ***
monthmar       2.105e-00  1.645e-01  12.799 < 2e-16 ***
monthmay       -4.582e-01  9.504e-02  -4.821 1.43e-06 ***
monthnov       -3.366e-01  1.149e-01  -2.929 0.003403 **
monthoct       3.625e-01  1.496e-01  2.423 0.015411 *
monthsep       6.097e-01  1.852e-01  3.292 0.000996 ***
poutcome nonexistent 5.038e-01  7.627e-02  6.605 3.97e-11 ***
poutcome success  8.023e-01  2.381e-01  3.369 0.000754 ***
emp.var.rate    -1.770e-00  1.881e-01 -10.528 < 2e-16 ***
cons.price.idx  2.551e-00  2.681e-01  9.517 < 2e-16 ***
contacttelephone -7.436e-01  9.276e-02  -8.016 1.09e-15 ***
cons.conf.idx   4.426e-02  6.544e-03  6.763 1.35e-11 ***
pdays         -1.127e-03  2.373e-04  -4.749 2.05e-06 ***
defaultunknown -3.873e-01  7.779e-02  -4.979 6.39e-07 ***
defaultyes      -7.220e-00  1.387e-02  -0.052 0.958487
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 20357 on 28831 degrees of freedom
Residual deviance: 12077 on 28811 degrees of freedom
AIC: 12119

Number of Fisher scoring iterations: 10

```

```

####r)
summary(backwardaicc_model)

call:
glm(formula = y_boolean ~ default + contact + month + duration +
  pdays + poutcome + emp.var.rate + cons.price.idx + euribor3m,
  family = "binomial", data = train)

Deviance Residuals:
    Min       1q   Median       3q      Max
-5.9718  -0.3027  -0.1861  -0.1382   3.1460

Coefficients:
              estimate std. error z value Pr(>|z|)
(Intercept)   -1.839e-02  1.181e-01 -15.576 < 2e-16 ***
defaultunknown -3.800e-01  7.780e-02  -4.884 1.04e-06 ***
defaultyes     -7.264e-00  1.388e-02  -0.052 0.958276
contacttelephone -6.291e-01  8.471e-02  -7.426 1.12e-13 ***
monthaug       9.299e-01  1.016e-01  9.153 < 2e-16 ***
monthdec       4.605e-01  2.194e-01  2.099 0.038835 *
monthjul       1.442e-01  1.091e-01  1.322 0.186193
monthjun       -4.422e-01  1.252e-01  -3.532 0.000412 ***
monthmar       1.910e-00  1.351e-01  14.137 < 2e-16 ***
monthmay       -5.134e-01  8.775e-02  -5.851 4.89e-09 ***
monthnov       -5.832e-01  1.270e-01  -4.591 4.41e-06 ***
monthoct       1.129e-01  1.391e-01  0.812 0.416906
monthsep       3.641e-01  1.384e-01  2.630 0.008531 **
duration       4.646e-03  8.799e-05  52.807 < 2e-16 ***
pdays        -1.134e-03  2.376e-04  -4.772 1.83e-06 ***
poutcome nonexistent 4.758e-01  7.613e-02  6.250 4.11e-10 ***
poutcome success  8.165e-01  2.386e-01  3.422 0.000621 ***
emp.var.rate    -1.784e-00  1.228e-01 -14.532 < 2e-16 ***
cons.price.idx  2.508e-00  1.232e-01  20.341 < 2e-16 ***
euribor3m       6.280e-01  9.360e-02  6.709 1.96e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 20357 on 28831 degrees of freedom
Residual deviance: 12085 on 28812 degrees of freedom
AIC: 12125

Number of Fisher scoring iterations: 10

```

In the end, the forward selection with BIC gives a 11-feature model with the lowest cross-validation MSE while the two AIC models give the lowest testing MSE.

```

####r)
print(paste0("Forward AIC 5-fold MSE: ", faic_mse))

[1] "Forward AIC 5-fold MSE: 0.141052782672364"

####r)
print(paste0("Backward AIC 5-fold MSE: ", baic_mse))

[1] "Backward AIC 5-fold MSE: 0.141328808424519"

####r)
print(paste0("Forward BIC 5-fold MSE: ", fbic_mse))

[1] "Forward BIC 5-fold MSE: 0.140532044629618"

####r)
print(paste0("Backward BIC 5-fold MSE: ", bbic_mse))

[1] "Backward BIC 5-fold MSE: 0.141251468466805"

```

```

####r)
print(paste0("Forward AIC testing MSE: ", faic.mse.test))

[1] "Forward AIC testing MSE: 0.0617030685717096"

####r)
print(paste0("Backward AIC testing MSE: ", baic.mse.test))

[1] "Backward AIC testing MSE: 0.0617030685717054"

####r)
print(paste0("Forward BIC testing MSE: ", fbic.mse.test))

[1] "Forward BIC testing MSE: 0.0621399608937084"

####r)
print(paste0("Backward BIC testing MSE: ", bbic.mse.test))

[1] "Backward BIC testing MSE: 0.0620861361469596"

```

However, the nature of forward/backward selection decides that maybe none of the three models is the actual optimal model. Also, the difference of the MSEs of the four models is very trivial. Some of

the remaining features of all three models are macro-economic features, including euribor3m, cons.price.idx, emp.var.rate. This makes sense because the interest rate, inflation rate, and employment rate are highly correlated with the market demand of bank savings by nature. The demographic profile of the potential customer does not seem to matter that much in the BIC models since age, job, marital status, and education level are not included in the two BIC models, while the AIC models kept some demographic variables. The most important demographic status is whether or not the customer has defaulted in the past. The marketing effort does matter since duration of the marketing call and the contact communication type can influence the success rate. However, we have to note that "duration" should only be included for benchmark purposes and should be discarded if the intention is to have a realistic predictive model. Customer loyalty also seems to play a role in the success rate of the marketing campaign since poutcome is in the model and thus, the success/failure of the last marketing campaign can influence the success rate of future campaign. The most surprising result is that the month of the year is important in determining the success rate. However, this might be due to the fact that this data is collected during the 2008-2010 financial crisis, and there might be several months in those three years that have particularly bad macro-economics conditions. Maybe customers in those months with "bad economies" might resort to or stay away from bank savings. This is worth further investigating in future researching.

Method 2: Shrinkage Methods

The two most used and useful shrinkage methods are the Lasso and Ridge regressions. These methods are used to reduce model complexity while preventing over-fitting which may happen when simple regressions are learned. The ridge regression adds a penalty proportional to the sum of the squares of the coefficient. This helps us reduce complexity and multi-collinearity. On the other hand, Lasso regression penalizes the sum of absolute values of the coefficients. This helps reduce over-fitting and helps with feature selection.

In terms of assumptions, both Lasso and ridge are parametric functions, so the linearity assumption needs to be met. However, both Lasso and Ridge are able to handle multicollinearity, so this assumption does not need to be met for our shrinkage methods. However, the independence and homoscedasticity assumptions should be met in order for the shrinkage methods to take full advantage of the regularizing terms.

We first benchmarked the generalized linear model without any regularizing terms to establish a baseline. Again, we are predicting the outcome of a banking marketing call using information about the individual and macroeconomic factors. Here are the results from the unregularized general linear model.

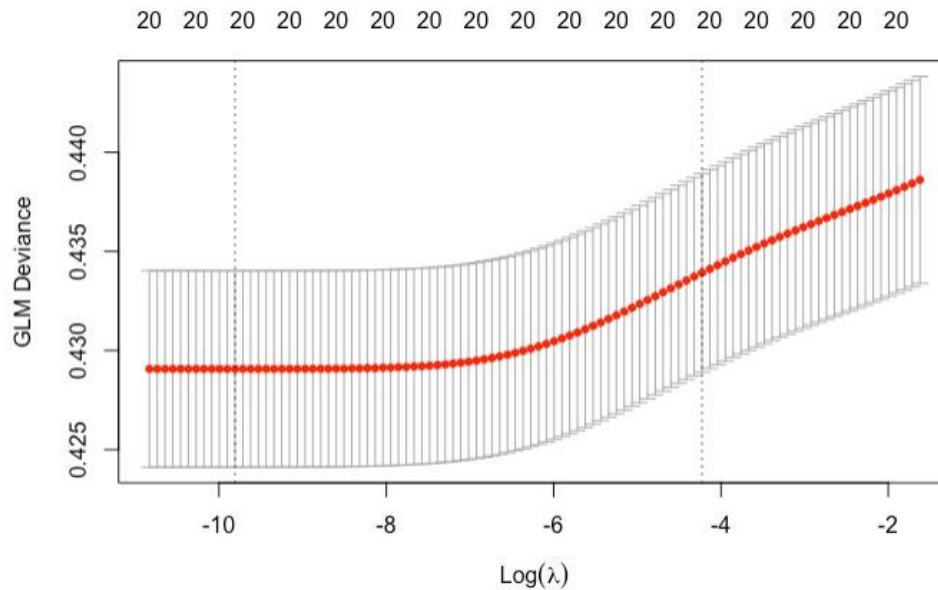
```
[1] "the unregularized MSE is: 0.0638723851269843"  
[1] "the unregularized accuracy is: 0.910246034315312"
```

And the intercepts are shown here.

(Intercept)	age	jobblue-collar	jobentrepreneur
-2.592084e+02	-5.188314e-04	-2.230757e-01	-1.543479e-01
jobhousemaid	jobmanagement	jobretired	jobself-employed
-1.330020e-01	-3.910358e-02	1.839547e-01	-1.270466e-01
jobservices	jobstudent	jobtechnician	jobunemployed
-1.110349e-01	1.574635e-01	-3.973814e-03	-1.252125e-01
jobunknown	maritalmarried	maritalsingle	maritalunknown
-2.089915e-01	3.440455e-02	4.207444e-02	1.514376e-01
educationbasic.6y	educationbasic.9y	educationhigh.school	educationilliterate
-1.315362e-02	-9.940191e-03	3.507052e-02	1.484119e+00
educationprofessional.course	educationuniversity.degree	educationunknown	defaultunknown
4.540590e-02	1.208174e-01	1.078687e-01	-3.288056e-01
defaultyes	housingunknown	housingyes	loanunknown
-7.295241e+00	-6.706570e-02	-1.776312e-02	NA
loanyes	contacttelephone	monthaug	monthdec
-8.332179e-02	-7.355614e-01	7.636090e-01	4.605033e-01
monthjul	monthjun	monthmar	monthmay
5.486555e-02	-5.818085e-01	1.991649e+00	-4.789203e-01
monthnov	monthoct	monthsep	day_of_weekmon
-5.174196e-01	1.574934e-01	4.113721e-01	-1.395910e-01
day_of_weekthu	day_of_weektue	day_of_weekwed	duration
2.131361e-02	8.364798e-02	1.997486e-01	4.651347e-03
campaign	pdays	previous	poutcomenonexistent
-3.826762e-02	-1.271723e-03	-1.136690e-01	3.566130e-01
poutcomesuccess	emp.var.rate	cons.price.idx	cons.conf.idx
6.783750e-01	-1.777293e+00	2.339730e+00	3.015945e-02
euribor3m	nr.employed		
2.543907e-01	7.327428e-03		

As we can see, all of the features are present in this regression, making inference somewhat difficult, and we do not have a clear understanding of what are the most important driving factors in predicting whether someone will be susceptible to marketing calls.

The first shrinkage method that was used is Ridge regression which penalized the squared sum of the coefficients. We created a custom sequence of lambdas to feed into the `cv.glmnet` function since there has been literature stating that the default values are not optimal. The graph below demonstrates the optimal log values of lambdas from the cross-validation that the package performs. Since we want the optimal MSE scores, we selected `lambda.min` value when building our optimal ridge regression model.



The accuracy measurements are shown here.

```
[1] "the ridge regression MSE is: 0.0633577130946119"
[1] "the ridge regression accuracy is: 0.911379087083198"
```

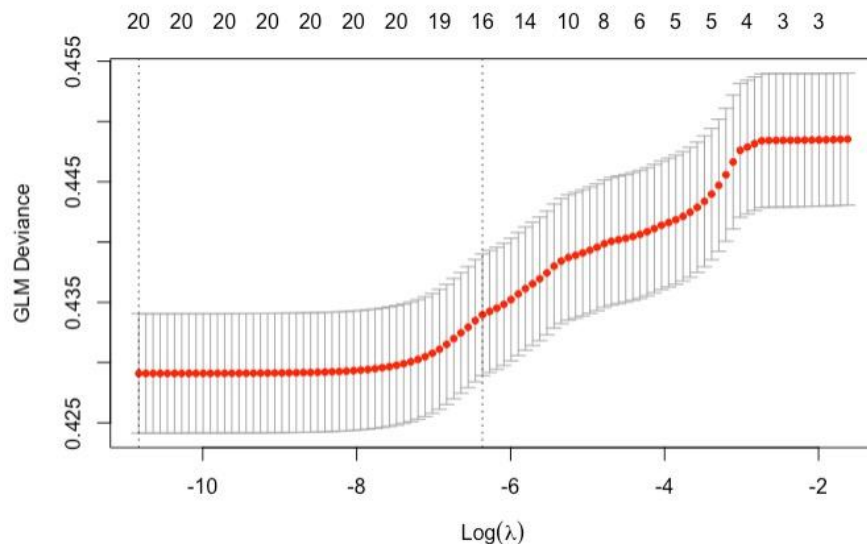
It has been previously shown that ridge regression should perform strictly not worse than the regularized version. And luckily our results confirm this. The ridge MSE is smaller than the 0.0638 value from the unregularized general linear model, and the ridge regression accuracy is also higher than the unregularized accuracy. This is perhaps because the penalty added in ridge regression helps reduce overfitting and gives us better testing accuracy.

	s1	day_of_week	0.057321509
(Intercept)	-16.110831784	duration	0.004527507
age	0.003996037	campaign	-0.030454284
job	0.005872729	pdays	-0.001253130
marital	0.081757932	previous	-0.131816215
education	0.042261375	poutcome	0.359423661
default	-0.397795178	emp.var.rate	-0.873639881
housing	-0.015310409	cons.price.idx	0.789834499
loan	-0.047068866	cons.conf.idx	0.028211614
contact	-0.747654334	euribor3m	0.536144337
month	-0.111748702	nr.employed	-0.011748028

Here are the coefficients that are from the ridge regression outputs. We see that compared to the unregularized version, the coefficients are significantly smaller (on the magnitude of 10x on average). However, none of the coefficients are zero, since the squared sum is penalized not the absolute value. One coefficient to highlight here is the poutcome coefficient, which suggests that if an individual has

committed before, they are more likely to commit again. Another coefficient to highlight is the contact variable's coefficient. Combined with poutcomes, we see that if an individual has been contacted before, but didn't commit, it's very unlikely that they will commit on the next call.

The last shrinkage method that was performed is the Lasso regression. The motivation here is similar to why the Ridge regression was used. We want to limit overfitting while attempting to understand the most important features of the predictions. The results are as follows:



In the graph above, we ran the same lambda schedule as for the ridge regression. Again, this is different than the default lambda sequence from `cv.glmnet`. We see that for larger log values of lambda, the deviance is higher, perhaps because the penalty is too large, and many important features are being truncated, causing a deterioration in performance from the model. Here, we select the right dotted vertical line or the `lambda.1se` value to be used, since we want to promote sparsity in the data.

```
[1] "the lasso regression MSE is: 0.0633616798599029"
[1] "the lasso regression accuracy is: 0.911136290061509"
```

By invoking the Lasso regression, the regression MSE also improves from the baseline no-shrinkage model. Perhaps this is from benefitting from less overfitting. We also see that the regression accuracy improves too, giving us more predictive power for our model.

	s1	day_of_week	0.044599896
(Intercept)	66.480254006	duration	0.004495338
age	0.001714778	campaign	-0.028404772
job	0.003878356	pdays	-0.001453730
marital	.	previous	-0.108295274
education	0.048865180	poutcome	0.179014172
default	-0.163906546	emp.var.rate	-0.086476972
housing	.	cons.price.idx	.
loan	.	cons.conf.idx	0.032732861
contact	-0.306069402	euribor3m	0.007578947
month	-0.098119270	nr.employed	-0.013072061

The coefficients from the model are shown above, where the dots indicate that the coefficient is zero, or in other words, the features are not included in the model. This behavior is expected from the Lasso shrinkage method. Furthermore, we see that marital status, housing, loan, and cons.price.idx are the features that are removed. It makes intuitive sense that marital status isn't very helpful in predicting whether someone will commit to a banking call, and we can understand the removal of cons.price.idx by the multicollinearity of the macroeconomic factors shown in the previous EDA report.

From the modeling portion, we see that we are able to predict whether an individual will commit to a banking marketing call with more than 91.1% accuracy, as that is the main objective of the model. Furthermore, we are able to see that marital status, housing, loan, and cons.price.idx are less significant when it comes to predictive power.

Method 3: Categorical variables / ANOVA

The main question we want to answer from this method is which factors are most important in predicting whether a client will subscribe to the term deposit or not. To answer this question, we will use a generalized linear model (GLM) with a binary response variable indicating whether a client subscribed to the term deposit or not. Before we build the GLM model, it is important to identify which features to include in the model in order to minimize the mean square error of our prediction. To do this, we will perform an analysis of variance (ANOVA) on the dataset to identify which features have a significant effect on the response variable. By including only the most significant features in our model, we can reduce the complexity of the model and improve its predictive power.

In this context, ANOVA provides a useful tool for identifying the features that have the greatest impact on the response variable. We will use the results of the ANOVA to guide our selection of features for the GLM model, allowing us to build a more accurate and interpretable model for predicting the likelihood of a client subscribing to the term deposit.

There are several assumptions associated with ANOVA that need to be checked to ensure the validity of the analysis.

- Independence: The observations should be independent of each other. In this dataset, each observation corresponds to a unique client, so this assumption is met.
- Normality: The residuals should be normally distributed. We can check this assumption by plotting a histogram of the residuals or by using a normal probability plot, but it was difficult to check this assumption directly on the bank-additional-full dataset because we don't have the actual predicted values to calculate the residuals.
- Homogeneity of variance: The variances of the different groups should be approximately equal. We can check this assumption by plotting a box plot of the residuals for each group.
- Independence of groups: The groups being compared should be independent of each other. In this dataset, the groups are independent because each client is assigned to only one group.

Overall, the bank-additional-full dataset seems to meet the assumptions required for ANOVA.

However, it is important to note that there may be other unmeasured factors that could impact the validity of the analysis.

In this analysis, we will be using the bank marketing dataset to perform an analysis of variance (ANOVA) to determine the significance of the relationship between various independent variables and the binary response variable 'y-binary'. The bank marketing dataset contains information on the marketing campaign of a Portuguese banking institution, including attributes of the clients, the marketing campaign, and the final outcome of the campaign.

First, we load the bank marketing dataset into R and convert the response variable 'y' to a binary numeric variable 'y-binary', with 1 representing a successful marketing campaign and 0 representing an unsuccessful campaign. Then, we use the 'aov' function in R to perform the ANOVA on 'y_binary' and all other independent variables in the dataset.

```
# Load the bank marketing dataset
bank_data <- read.csv("~/Desktop/151 final proj/bank-additional-full.csv", sep=";")

# Convert the response variable y to a binary numeric variable
bank_data$y_binary <- ifelse(bank_data$y == "yes", 1, 0)

#anova model
anova_model <- aov(y_binary ~ ., data = bank_data)

summary(anova_model)
```

```
##          Df Sum Sq Mean Sq  F value Pr(>F)
## age          1    3.8      3.8 4.246e+31 <2e-16 ***
## job         11   92.8      8.4 9.415e+31 <2e-16 ***
## marital       3    8.1      2.7 2.999e+31 <2e-16 ***
## education     7    5.8      0.8 9.185e+30 <2e-16 ***
## default       2   31.2     15.6 1.739e+32 <2e-16 ***
## housing       2    0.3      0.1 1.604e+30 <2e-16 ***
## loan          1    0.2      0.2 1.748e+30 <2e-16 ***
## contact       1   57.7     57.7 6.437e+32 <2e-16 ***
## month         9  221.8     24.6 2.750e+32 <2e-16 ***
## day_of_week   4    4.1      1.0 1.131e+31 <2e-16 ***
## duration      1  639.7    639.7 7.139e+33 <2e-16 ***
## campaign      1    0.7      0.7 7.380e+30 <2e-16 ***
## pdays         1  225.7    225.7 2.518e+33 <2e-16 ***
## previous      1    0.5      0.5 5.086e+30 <2e-16 ***
## poutcome      2    4.2      2.1 2.325e+31 <2e-16 ***
## emp.var.rate  1  102.2    102.2 1.141e+33 <2e-16 ***
## cons.price.idx 1   53.7     53.7 5.994e+32 <2e-16 ***
## cons.conf.idx  1   17.2     17.2 1.917e+32 <2e-16 ***
## euribor3m     1    4.9      4.9 5.496e+31 <2e-16 ***
## nr.employed   1    0.0      0.0 2.655e+29 <2e-16 ***
## y             1 2642.9   2642.9 2.949e+34 <2e-16 ***
## Residuals    41134    0.0      0.0
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA results are shown in the table above. The ANOVA table shows the results of testing the significance of each variable in the model, including age, job, marital status, education, default, housing, loan, contact, month, day of week, duration, campaign, pdays, previous, poutcome, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed, and the response variable y. Each row represents an independent variable, with columns showing the degrees of freedom (Df), the sum of squares (Sum Sq), the mean sum of squares (Mean Sq), the F-value, and the p-value (Pr(>F)).

We can see that all independent variable, except for 'nr.employed' and 'loan', have extremely low p-values (less than 2e-16), indicating that they have a significant relationship with the response variable 'y_binary'. This suggests that these variables may be important predictors of the success of a marketing campaign.

Furthermore, the F-values for each variable are extremely large, indicating a large amount of variance in the response variable that can be explained by each independent variable. For example, the variable 'duration' has an F-value of 7.139e+33, indicating that it explains a large amount of the variance in 'y_binary'. It should be noted that including the feature 'duration' in the model may only serve as a benchmark for comparison, and it may not be appropriate to include it in a realistic predictive model.

Overall, this ANOVA analysis suggests that there are several independent variables in the bank marketing dataset that are significantly related to the success of a marketing campaign. These variables include age, job, marital status, education, default, housing, contact, month, day of week, duration, campaign, pdays, previous, poutcome, emp.var.rate, cons.price.idx, cons.conf.idx, and euribor3m.

It is important to note that these variables may be correlated with each other, which could impact the results of any predictive model. Further analysis, such as correlation analysis or regression analysis, could be performed to investigate these relationships in more detail.

In conclusion, this ANOVA analysis of the bank marketing dataset has identified several independent variables that are significantly related to the success of a marketing campaign. These variables should be considered when developing a predictive model for marketing campaign success. However, it is important to be mindful of potential correlations between these variables and to perform further analysis to investigate these relationships.

Conclusion

From using ANOVA, we were able to conclude that the age, job, marital status, education, default, housing, contact, month, day of week, duration, campaign, pdays, previous, poutcome, emp.var.rate, cons.price.idx, cons.conf.idx, and euribor3m variables are all significant predictors for our outcome variable. This suggests that the outcome of banking call success cannot be simply reduced to one or two variables that capture most of the variance. As a next step, we used shrinkage methods to attempt to truncate the number of features. The ridge regression and lasso regressions were both helpful in reducing overfit and helped increased testing accuracy while reducing MSE. From the outputs of the cross-validated Lasso output, we see that the marital status, housing, loan, and cons.price.idx are not as helpful to achieve higher accuracy and lower MSE.

Lastly, and more aggressively, we used model selection from AIC/BIC scores to build modeling in a stepwise manner. It is hard to determine the best model since their MSEs are really close, but the most aggressive model that was created from the model selection process was from the backward BIC models, and some remaining features are poutcome, month, pday, default, contact, euribor3m, cons.price.idx, and emp.var.rate, while the forward/backward AIC models have the lowest testing MSE, and the forward BIC model has the lowest cross-validation MSE within the training set. This method may not have produced the most optimal combination of features, but helps us understand the relative importance of the features regardless.

By utilizing these modeling tools, we understand that the prediction of banking marketing call success is embedded in a relatively high dimension, and cannot be simply reduced to a few principle features that help with prediction. However, through various trials and error, we were able to achieve 91% accuracy with the modeling tools.

A rather large limitation that we encountered is that the outcome variable is binary in nature. Even though we used a binomial family with a logit link function, we are still not perfectly capturing the nature of the underlying data. We would like to explore other techniques such as decision trees in order to make the models more interpretable while being able to handle classification tasks more efficiently. Our results from the model selection part yielded the conclusion that maybe customers in those months with "bad economies" might resort to or stay away from bank savings. An interesting future study that could be

inspired by this is a to create a calling schedule that takes the individuals' information into account while factoring into the macroeconomic conditions. Perhaps banks can leverage these calling schedules to encourage more people to commit to banking calls.